Ankit Das

AI Engineer | Computer Vision, Document AI and Machine Learning
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SUMMARY

AI engineer who ships machine learning into products. At IDVerse (LexisNexis Risk Solutions), built document-understanding systems across 10+ document types (OCR, computer vision, VLM/LLM inference, LoRA/QLoRA fine-tuning) and deployed them as Python FastAPI/Docker services, including an AWS EC2 service piloted with real users. Builds evaluation and regression gates so model changes are measured before release.

EXPERIENCE

IDVerse / LexisNexis Risk Solutions (RELX) | Feb 2025 – Aug 2026 | Remote
Computer Vision Engineer

- Built document-understanding pipelines for 10+ identity and utility document types, combining OCR, layout and geometry signals, image preprocessing, VLM/LLM inference, structured extraction and field validation.
- Fine-tuned models with LoRA/QLoRA and added retrieval-assisted extraction; designed and built an AWS EC2 extraction service around an open-source vision-language model, piloted with real users.
- Trained multilingual OCR models (CNN + bidirectional LSTM + CTC) for 6 scripts, quantized to ~9 MB INT8 ONNX models for CPU and in-browser execution in a proof-of-address extraction SDK.
- Built a 994-document, 7-language dataset and key-information extraction models (BiGRU-CRF, layout-aware Transformer) that separate the customer's address and issue date from look-alike text.

CybraneX | Oct 2023 – Mar 2024 | Remote
Quantum Machine Learning Intern

- Formulated portfolio optimization as a QUBO/Ising model solved with Monte Carlo simulated annealing; contributed to TrOCR/Azure ML text-extraction experiments; helped coordinate a 10+ intern cohort.

PROJECTS

MP-KG-RAG | Graph-based RAG with citation validation
Python, Qwen, BGE-M3 embeddings, knowledge graphs, structured generation

- Built an end-to-end RAG pipeline: hashed ingestion, sentence-aligned chunking, knowledge-graph retrieval, five perspective analyses and schema-constrained LLM synthesis with citations, automatic repair and quarantine.
- Ran four controlled variants (zero-shot, few-shot, KG-RAG, multi-perspective KG-RAG) over 6,200 outputs, with citation checks and multilingual evaluation in English, Hindi and Tamil (chrF, BERTScore).

Name That LLM | Kaggle competition | github.com/PiKa919/kaggle-name-that-llm
Python, PyTorch, Hugging Face, LoRA, cross-validation

- Built an 18-class classifier that identifies which LLM wrote a text: six LoRA-tuned backbones plus stylometric features, prompt-grouped cross-validation against topic leakage, ablations and out-of-fold fusion.
- Best public leaderboard score 0.661; cut feature memory to fit Kaggle's 30 GB limit. 8 Kaggle competitions completed overall (best: 532 / 4,381 in rainfall prediction).

ForgeSight | Model serving with evaluation gates | github.com/PiKa919/forgesight
Python, FastAPI, PyTorch, ONNX Runtime, PostgreSQL

- Added model and data provenance, automated evaluation and regression gates for releases, plus latency tracing; ONNX Runtime delivered 1.47x PyTorch throughput on an Apple M5 CPU.

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript
Generative AI: RAG, embeddings, knowledge graphs, LLM/VLM inference, LoRA/QLoRA fine-tuning, structured outputs, prompt design
ML and vision: PyTorch, Hugging Face, ONNX Runtime, OCR, computer vision, pandas, NumPy, cross-validation, ablations
Platforms: AWS (EC2, S3, Lambda), Azure ML (internship), Docker, FastAPI, PostgreSQL, Git
Certification: AMD ROCm Certified Associate (2026)

PUBLICATION

Co-author, "Quantum Optimization Algorithms and Application" (QAOA, VQE, quantum annealing, Grover's search), Chapter 6, pp. 91–120, in Quantum Computing Unleashed: Design, Models and Applications (Scrivener / Wiley, 2026).

EDUCATION

B.Tech, Computer Engineering with Honours in Data Science | 2021 – 2025
Shah & Anchor Kutchhi Engineering College, University of Mumbai | CGPA 8.05/10